

Estimating Nonlinear Insulin-Glucose Dynamics in Critically-ill Patients

Markus Mulvihill

KTH Royal Institute of Technology

August 7th, 2026



2026-08-26

Thesis Presentation

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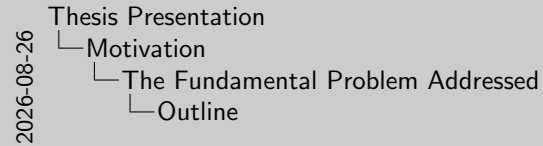
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1. Supervisor - Martin Lindström
2. Examiner - Ragnar Thobaben
3. Opponent - Divyayan Day

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 - The Fundamental Problem Addressed
 - Background
 - Previous Work
- 2 Results and Contributions
 - Main Results
 - Key Ideas Behind Implementation
- 3 Summary
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Fundamentals of Glucoregulation

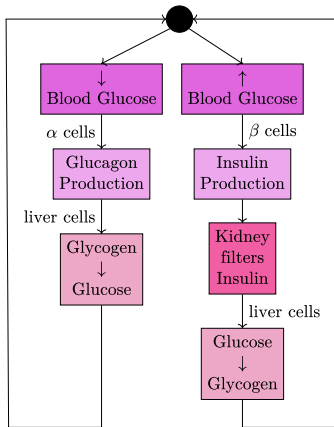


Figure: Flowchart of glucoregulation in healthy individuals.

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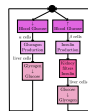
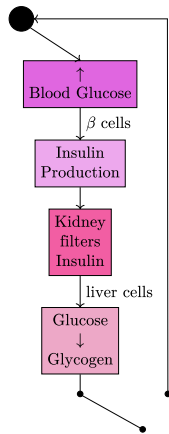


Figure: Flowchart of glucoregulation in healthy individuals.

1. The thesis revolves around the functionality of glucoregulation.
2. A high level diagram of the human insulin-glucose system can be described as the image.

Insulin Resistance in ICU Patients



- Critically ill patients' bodies do not respond normally to insulin.

Figure: Consequence of insulin resistance in ICU patients.

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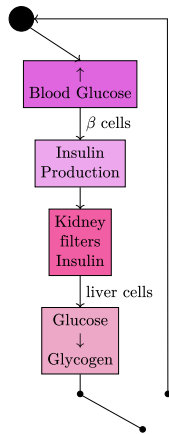
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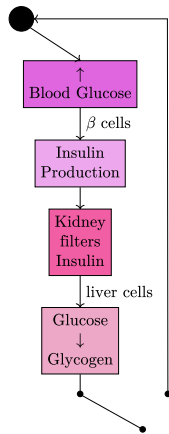


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- Impaired glycogen production in the liver creates an excess of blood glucose (hyperglycaemia).

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Insulin Resistance in ICU Patients

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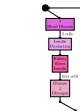


Figure: Consequence of insulin resistance in ICU patients.

- Critically ill patients' bodies do not respond normally to insulin.
- Stress induced from critical illness has been observed with an increase of resistance to insulin.
- Impaired glycogen production in the liver creates an excess of blood glucose (hyperglycaemia).

Mention that the break in the chain is just for visual purposes.

Intensive Insulin Therapy

- **What:**
- **Goal:**
- **How:**

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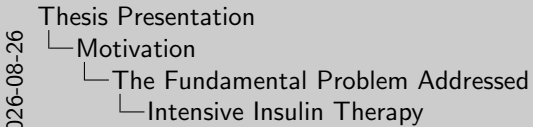
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Motivation

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Intensive Insulin Therapy

- **What:** Utilizes aggressive insulin infusions and feeding inputs to tightly control blood glucose.
- **Goal:**
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- **How:**

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- **How:** Traditionally has relied on ad-hoc sliding protocols.

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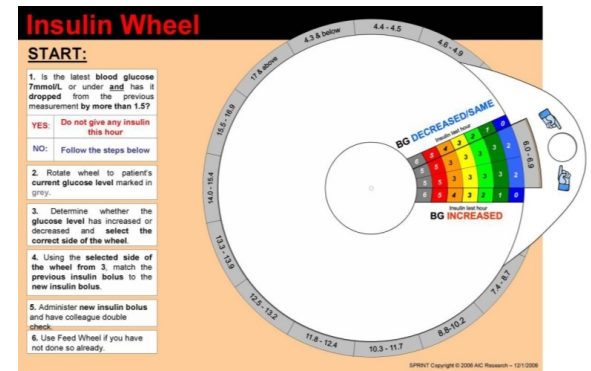
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Overview of Ad-hoc Sliding Protocols



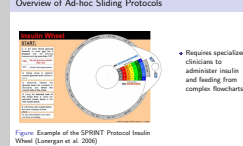
- Requires specialized clinicians to administer insulin and feeding from complex flowcharts.

Figure: Example of the SPRINT Protocol Insulin Wheel (Loneragan et al. 2006)

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Overview of Ad-hoc Sliding Protocols

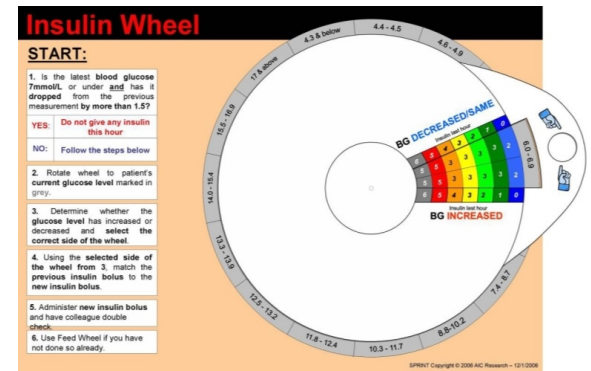


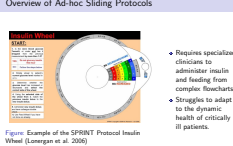
Figure: Example of the SPRINT Protocol Insulin Wheel (Loneragan et al. 2006)

- Requires specialized clinicians to administer insulin and feeding from complex flowcharts.
- Struggles to adapt to the dynamic health of critically ill patients.

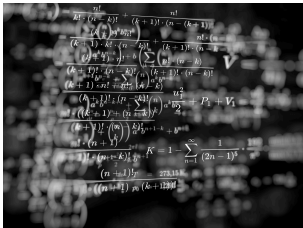
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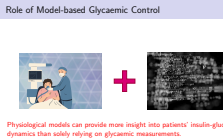
Role of Model-based Glycaemic Control



Physiological models can provide more insight into patients' insulin-glucose dynamics than solely relying on glycaemic measurements.

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 - Role of Model-based Glycaemic Control



Insulin infusions combined with better understanding of dynamics can provide more insight to the clinicians

1 Motivation

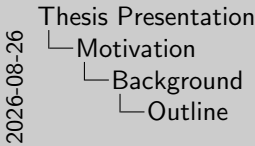
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Intensive Control Insulin-Nutrition-Glucose (ICING) model

Lin et al. has developed a clinically-validated nonlinear insulin-glucose model with the following main components

Dynamic	Notation	Value
Blood Glucose	$\dot{G}$	mmol/L
Interstitial Plasma	$\dot{Q}$	mU/L
Plasma Insulin	$\dot{i}$	mU/L
Glucose in the Stomach	$\dot{P1}$	mmol
Glucose in the Gut	$\dot{P2}$	mmol
Endogenous Insulin Production	$u_{en}(t)$	mU/min
Insulin Sensitivity	$S_I(t)$	-
Input	Notation	Value
Exogenous Insulin	$u_{ex}(t)$	mU/min
External Food	$D(t)$	mmol/min
Parenteral Glucose	$P_N(t)$	mmol/min

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Computational Approximation of the ICING model

- The ICING model equations are defined in continuous-time
- Discretization of the model is required for the presented estimation methods

For an example dynamic F

$$\begin{aligned}t_k &= t_{k-1} + \Delta \\ \dot{F} &\approx \frac{F(t_k) - F(t_k - \Delta)}{\Delta} \\ F_k &\approx F_{k-1} + \Delta \dot{F}.\end{aligned}$$

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1. According to Euler's method
2. Discretization error is bounded by Δ

The discrete-time dynamic ICING model can be represented in the stochastic state-space.

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 - Bayesian Filtering

Bayesian Filtering

The discrete-time dynamic ICING model can be represented in the stochastic state-space.

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State-space Representation

$$x_k \sim p(x_k \mid x_{k-1}, u_{k-1})$$
$$y_k \sim p(y_k \mid x_k)$$

System Variable	Notation
Dynamics	x_k
Inputs	u_k
Measurements	y_k

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1. The measurement often depends on the hospital, but will be discussed more later

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There are various methods of optimal filtering, and each has their own assumptions and limitations.

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There are various methods of optimal filtering, and each has their own assumptions and limitations.

1. The measurement often depends on the hospital, but will be discussed more later
2. Just the EKF and the PF will be in the topic of this thesis

Extended Kalman Filter

Approximates **nonlinear** and **Gaussian** dynamic and measurement models as a linear transformation.

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Extended Kalman Filter

Approximates **nonlinear** and **Gaussian** dynamic and measurement models as a linear transformation.

1 Prediction step

$$\begin{aligned}m_k^- &= f(m_{k-1}, u_{k-1}), \\P_k^- &= F_x(m_{k-1})P_{k-1}F_x^T(m_{k-1}) + F_q(m_{k-1})Q_{k-1}F_q^T(m_{k-1}).\end{aligned}$$

2 Update step

$$\begin{aligned}v_k &= y_k - h(m_k^-), \\S_k &= H_x(m_k^-)P_k^-H_x^T(m_k^-) + H_r(m_k^-)R_kH_r^T(m_k^-), \\K_k &= P_k^-H_k^T(m_k^-)S_k^{-1}, \\m_k &= m_k^- + K_kv_k, \\P_k &= P_k^- - K_kS_kK_k^T.\end{aligned}$$

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Bootstrap Particle Filter

Formulates the posterior of the dynamic function as a Monte Carlo approximation.

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Bootstrap Particle Filter

Formulates the posterior of the dynamic function as a Monte Carlo approximation.

- 1 Sample N particles from the dynamic function

$$x_k^{(i)} \sim p(x_k \mid x_{k-1}^{(i)}, u_{k-1}) \quad i = 1, \dots, N.$$

- 2 Update weights from the measurement

$$w_k^{*(i)} \propto p(y_k \mid x_k^{(i)}) \quad i = 1, \dots, N,$$
$$w_k^{(i)} = \frac{w_k^{*(i)}}{\sum_{j=1}^N w_k^{*(j)}} \quad i = 1, \dots, N.$$

- 3 Systematically resample particles from a distribution proportional to the weights

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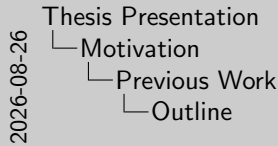
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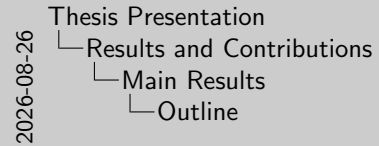
Formulation of Synthetic Patients

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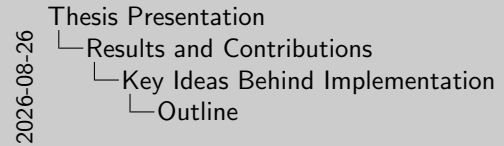
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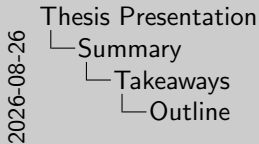
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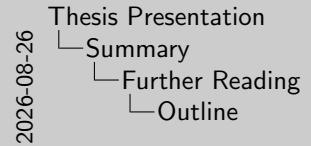
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 [Lin, Jessica et al. \(May 2011\). "A physiological Intensive Control Insulin-Nutrition-Glucose \(ICING\) model validated in critically ill patients". In: DOI: 10.1016/j.cmpb.2010.12.008.](#)

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